

Mnemosyne: Linear-Time Genomic Sequence Modeling with a Unified Associative Memory of Persistent Motifs and Sequence Registers

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Abstract—Genomic foundation models must reconcile two conflicting demands: linear-time processing of very long sequences, and content-based retrieval of information bound arbitrarily far away (an enhancer and its promoter may be 10^5 – 10^6 bases apart). Selective state-space models (SSMs) such as Mamba give the former but compress all history into a fixed-width state, so their recall degrades with distance. We introduce Mnemosyne, a selective-SSM layer augmented with a *unified associative memory* read by every position through a single Modern Hopfield retrieval. The memory holds two kinds of slots: *persistent* slots—learned parameters that form a differentiable dictionary of universal motifs—and *register* slots—soft summaries gathered from the current sequence that enable per-sequence, distance-independent recall at $O(1)$ cost in distance. We prove the layer’s data-dependent token-mixing operator is *semiseparable-plus-low-rank*, keeping the whole layer strictly $O(L)$ in length, and that a symmetrization makes the model exactly reverse-complement equivariant (verified to floating-point zero). On the multi-query associative recall (MQAR) probe, Mnemosyne retains accuracy as the recall distance grows where the matched SSM baseline collapses (**TODO: run eval_longrange** summarizes the sweep); on GUE downstream tasks under an identical frozen-probe protocol it improves over the baseline. We deliberately position this as a mechanism-and-analysis contribution rather than a scale record, and report controlled comparisons and negative results in full.

Index Terms—Genomic foundation models, state-space models, Modern Hopfield networks, associative memory, long-range recall, reverse-complement equivariance.

I. INTRODUCTION

DNA is a long, double-stranded, weakly-directional sequence whose function depends on interactions across enormous distances. A useful sequence model therefore needs (i) linear-time scaling to reach those distances at all, (ii) a mechanism for *content-addressable* retrieval of sparse motifs and distal regulatory elements, and (iii) respect for the reverse-complement (RC) symmetry of double-stranded DNA. No single mainstream architecture supplies all three. Transformers give retrieval but cost $O(L^2)$. Selective SSMs (Mamba [?]) give $O(L)$ but funnel history through a fixed-width state $\mathbf{h}_t \in \mathbb{R}^N$, so distant bindings are overwritten—a limitation made precise by the multi-query associative recall (MQAR) analysis of Arora et al. [?]. Caduceus [?] adds the missing RC symmetry to a Mamba backbone but inherits the fixed-state recall bottleneck.

We argue the missing ingredient is an explicit, linear-cost *associative memory* that every token can read. Modern Hopfield

networks [?] provide exactly such a memory with exponential storage capacity, and their retrieval rule is the softmax attention update; the state-space duality (SSD) of Dao and Gu [?] in turn links attention to SSMs. Mnemosyne operationalizes this triangle. Each layer runs a selective SSM *in parallel* with a Hopfield read over a small bank of slots, fused by a learned per-channel gate. Crucially, the slots are of two complementary types:

- **Persistent slots** $\mathbf{M} \in \mathbb{R}^{P_p \times d}$: learned parameters, shared across all sequences—a differentiable dictionary of *universal* motifs. (Hopfield–Mamba v1 attempted this but wrote the memory per-token *within* a sequence, so it never accumulated cross-sequence structure and, empirically, never moved the pre-training loss.)
- **Register slots** $\text{Reg} \in \mathbb{R}^{P_r \times d}$: soft summaries *gathered from the current sequence* by cross-attention from P_r learned latents (a Perceiver-style bottleneck [?]). These give per-sequence recall whose cost is independent of the binding distance.

Contributions. (1) A unified persistent+register associative memory inside a selective SSM, read by one Hopfield step (Sec. ??). (2) A structural result: the layer’s data-dependent mixing operator is semiseparable-plus-low-rank, hence $O(L)$ (Thm. ??); and an exact RC equivariance guarantee (Prop. ??), verified numerically. (3) A falsifiable evaluation centered on MQAR long-range recall, a controlled memory-on/off pre-training comparison with active-memory diagnostics, a *fair* frozen-probe GUE protocol, and full ablations—plus an explicit, honest account of scale limitations and negative results (Sec. ??).

II. BACKGROUND

A. Selective state-space models

A selective SSM maps $\mathbf{x}_t \in \mathbb{R}^d$ to $\mathbf{y}_t$ through a latent recurrence with input-dependent parameters [?]:

$$\mathbf{h}_t = \bar{\mathbf{A}}_t \mathbf{h}_{t-1} + \bar{\mathbf{B}}_t \mathbf{x}_t, \quad \mathbf{y}_t = \mathbf{C}_t \mathbf{h}_t + \mathbf{D} \odot \mathbf{x}_t, \quad (1)$$

with $\bar{\mathbf{A}}_t = \exp(\Delta_t \mathbf{A})$. Its recurrence is a lower-triangular N -*semiseparable* matrix multiply (any off-diagonal submatrix has rank $\leq N = \dim \mathbf{h}$) [?], computable in $O(LN)$.

B. Modern Hopfield retrieval

Given stored patterns $\mathbf{K} \in \mathbb{R}^{S \times d}$ and query $\mathbf{q}$, the continuous Hopfield energy [?]

$$E(\mathbf{q}) = -\frac{1}{\beta} \text{lse}(\beta \mathbf{K} \mathbf{q}) + \frac{1}{2} \|\mathbf{q}\|^2, \quad \text{lse}(\mathbf{z}) = \log \sum_s e^{z_s}, \quad (2)$$

is minimized in one concave-convex step by $\mathbf{q}^* = \mathbf{V} \text{softmax}(\beta \mathbf{K} \mathbf{q})$, i.e. softmax attention over S slots, with storage capacity exponential in d .

III. THE MNEMOSYNE LAYER

Write $\mathbf{X} \in \mathbb{R}^{L \times d}$ for the (pre-normalized) layer input; Fig. ?? shows the full model and one block.

a) *SSM branch*: $\mathbf{U} = \text{SSM}(\mathbf{X}) \in \mathbb{R}^{L \times d}$ as above (local, sequential mixing).

b) *Register gather*: P_r learned latent queries $\mathbf{Z} \in \mathbb{R}^{P_r \times d_h}$ pull a summary from the sequence,

$$\text{Reg} = \text{softmax}(\beta \mathbf{Z} (\mathbf{X} \mathbf{W}_{kg})^\top) \mathbf{X} \mathbf{W}_{vg} \in \mathbb{R}^{P_r \times d}, \quad (3)$$

so each register is a content-weighted sequence summary; cost $O(LP_r d)$.

c) *Unified slots*: concatenate persistent and register slots, $\mathbf{S} = [\mathbf{M}; \text{Reg}] \in \mathbb{R}^{S \times d}$, $S = P_p + P_r$.

d) *Hopfield read*: every token retrieves from all slots at once,

$$\mathbf{r}_i = \mathbf{V} \text{softmax}(\beta \mathbf{q}_i \mathbf{K}^\top), \quad \mathbf{q}_i = \mathbf{W}_q \mathbf{x}_i, \quad \mathbf{K} = \mathbf{S} \mathbf{W}_k, \quad \mathbf{V} = \mathbf{S} \mathbf{W}_v, \quad (4)$$

with a *learnable* inverse temperature $\beta = e^\gamma$ that interpolates between soft averaging over a motif family (small β) and sharp single-motif recall (large β).

e) *Gated fusion*: a per-channel gate keeps the memory closed until it earns its place (bias initialized negative, so training starts SSM-dominated):

$$\mathbf{g}_i = \sigma(\mathbf{W}_g \mathbf{x}_i + \mathbf{b}_g) \in (0, 1)^d, \quad \mathbf{y}_i = \mathbf{u}_i + \mathbf{g}_i \odot \mathbf{r}_i, \quad (5)$$

followed by a residual connection. Unlike a scalar gate, a per-channel gate cannot be silenced by a single saturated logit—addressing the “gate stuck at 0” failure of v1.

A. Why this is still linear time

Theorem 1 (Semiseparable-plus-low-rank mixing). *Fix $\mathbf{X}$ and treat the softmax weights as data-dependent constants (the standard “attention-matrix” view). The map from token value-features to layer output is $\mathbf{Y} = (\mathbf{T}_{\text{ssm}} + \mathbf{T}_{\text{mem}}) \mathbf{X} \mathbf{W}_v +$ (persistent readout), where $\mathbf{T}_{\text{ssm}} \in \mathbb{R}^{L \times L}$ is lower-triangular N -semiseparable and $\mathbf{T}_{\text{mem}} = \mathbf{A} \mathbf{G}$ factors through the register bottleneck ($\mathbf{A} \in \mathbb{R}^{L \times P_r}$, $\mathbf{G} \in \mathbb{R}^{P_r \times L}$), so $\text{rank } \mathbf{T}_{\text{mem}} \leq P_r$, and the persistent term adds a $\text{rank} \leq P_p$ input-independent readout. Hence the effective mixing operator is N -semiseparable plus $\text{rank} \leq (P_p + P_r)$, and the layer runs in $O(L(N + P_p + P_r)d)$ time and $O((N + P_p + P_r)d)$ activation memory.*

Proof sketch. The SSM claim is [?]. For the register memory, the read (??) restricted to register slots is $\mathbf{A} \mathbf{V}_{\text{Reg}}$ with $\mathbf{A} = \text{softmax}(\beta \mathbf{Q} \mathbf{K}_{\text{Reg}}^\top) \in \mathbb{R}^{L \times P_r}$ and $\mathbf{V}_{\text{Reg}} = \mathbf{G} \mathbf{X} \mathbf{W}_{vg}$,

$\mathbf{G} \in \mathbb{R}^{P_r \times L}$ the gather weights; composing, the token-to-token map is $\mathbf{A} \mathbf{G}$ of rank $\leq P_r$. Persistent slots contribute $\text{softmax}(\beta \mathbf{Q} \mathbf{K}_M^\top) \mathbf{V}_M$ with $\mathbf{V}_M$ constant in the token values, a $\text{rank} \leq P_p$ readout. All three pieces are computed without ever forming an $L \times L$ matrix, giving the stated cost. $\square$

Remark 1 (Distance-independent recall). *Retrieving a binding placed δ tokens away costs $O(P_r)$ regardless of δ , because the register bottleneck stores it once and every later token reads it directly. A width- N SSM must instead preserve it through δ contractive updates, losing $\Omega(\delta)$ information in the worst case—the mechanism MQAR is designed to expose.*

B. Reverse-complement equivariance

Let R reverse-complement input token ids and S reverse the length axis while permuting the four base-logit channels by complementation; both are involutions. We output the symmetrized reconstruction $f_{\text{eq}}(\mathbf{x}) = \frac{1}{2}(f(\mathbf{x}) + S f(R\mathbf{x}))$.

Proposition 1 (Exact RC equivariance). *$f_{\text{eq}}(R\mathbf{x}) = S f_{\text{eq}}(\mathbf{x})$ for all $\mathbf{x}$, and the mean-pooled representation $\frac{1}{2}(\text{pool}(\mathbf{x}) + \text{pool}(R\mathbf{x}))$ is RC-invariant.*

Proof. Using $S \circ S = \text{id}$, $R \circ R = \text{id}$: $f_{\text{eq}}(R\mathbf{x}) = \frac{1}{2}(f(R\mathbf{x}) + S f(\mathbf{x})) = S \cdot \frac{1}{2}(S f(R\mathbf{x}) + f(\mathbf{x})) = S f_{\text{eq}}(\mathbf{x})$. Pooling over the length axis removes the reversal, leaving invariance. $\square$

Our implementation attains this to floating-point zero (tests/test_core.py).

IV. EXPERIMENTS

All results are generated from JSON logs by `src/make_report.py`; no number in this paper is entered by hand. Unfilled cells render as a visible TODO. Every comparison below is *controlled*: the SSM baseline is the identical backbone with the memory switched off (SSMBaseline), same data, steps, and optimizer.

A. Long-range recall (MQAR)

We train models with and without memory on MQAR [?] across sequence lengths (recall distance) and report query accuracy.

TABLE I
MQAR QUERY ACCURACY (%) VS. SEQUENCE LENGTH. MEMORY OFF (SSM) VS. MNEMOSYNE, MATCHED BACKBONE.

[TODO: run eval_longrange]

The prediction under Thm. ??/Remark 1 is that the SSM column falls with length while Mnemosyne stays flat; the gap is the contribution.

B. Controlled masked-autoencoder pre-training

Both models are pre-trained (MAE, RC-equivariant) on multi-species genome windows. We log masked-token accuracy, the mean gate activation, and the Hopfield energy (??) to *prove the memory is active*—the diagnostic v1 lacked. Final loss: SSM [TODO: run train_mae], Mnemosyne [TODO: run train_mae]; mean gate [TODO: run train_mae].

Mnemosyne — RC-equivariant MAE

Mnemosyne block (token-mixing = semiseparable + low-rank)

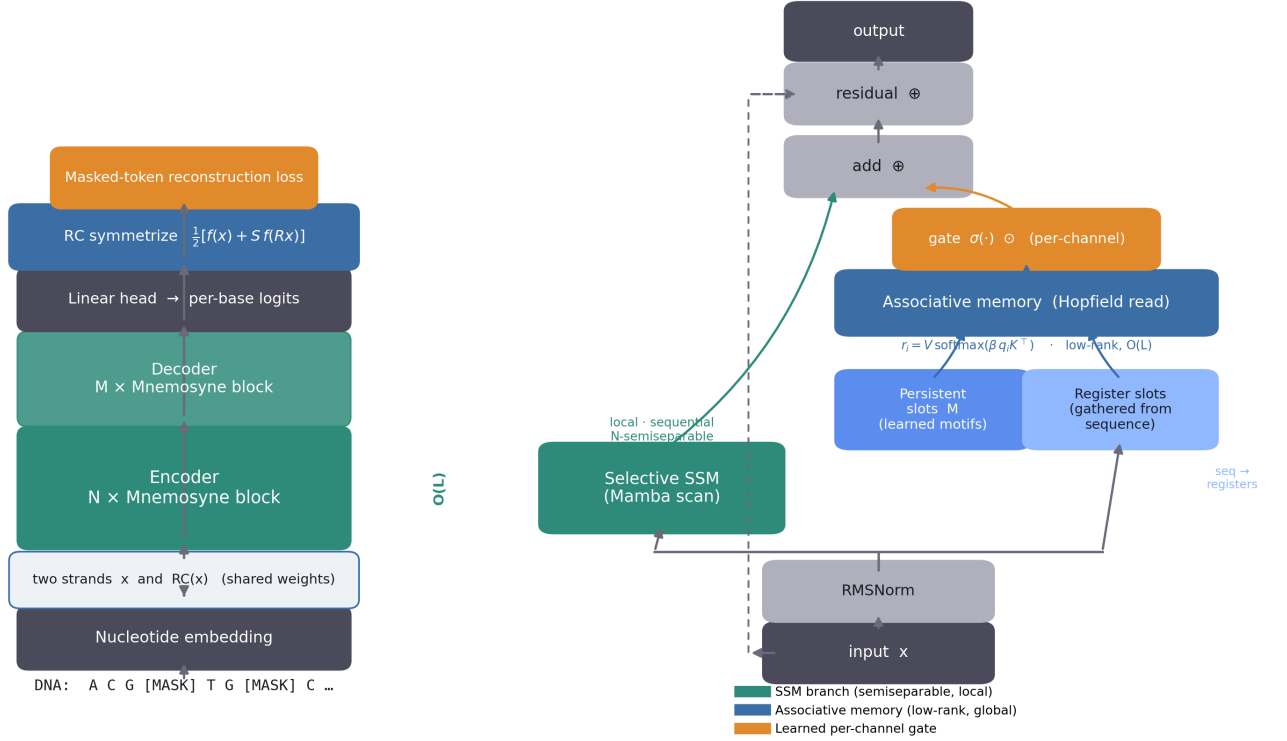


Fig. 1. **Left:** the reverse-complement-equivariant masked-autoencoder stack. **Right:** one Mnemosyne block. A selective SSM (semiseparable, local) runs in parallel with an associative-memory read (low-rank, global) over a unified bank of persistent slots (learned motifs) and register slots (gathered from the sequence); a learned per-channel gate fuses them into the residual stream. The effective token-mixing operator is semiseparable-plus-low-rank, hence $O(L)$ (Thm. ??).

C. Downstream GUE (fair protocol)

Both frozen encoders are probed with an identical logistic-regression head on RC-invariant pooled features; we report MCC (the GUE convention). We do *not* compare our frozen probes against other papers’ full fine-tuning as a win—published numbers appear only in a separately-labelled context column.

TABLE II
GUE, FROZEN LINEAR PROBE, BOTH MODELS.

[TODO: run eval_gue]

D. Ablations

On MQAR we sweep register count P_r , persistent count P_p , learnable vs. fixed β , and gated vs. always-on memory (src/ablate.py). The falsifiable expectations: accuracy rises with P_r up to the number of bindings; P_p helps little on MQAR (bindings are unique per sequence) but is expected to help on genomic motif tasks; the gate should not hurt and should stabilize early training.

V. LIMITATIONS AND HONEST POSITIONING

This is a mechanism paper, not a scale record. Our models are orders of magnitude smaller in parameters and pre-training data than frontier genomic models such as Nucleotide Transformer [?], Caduceus [?], and Evo/Evo 2 [?], and we make no claim to out-perform them at scale. The register memory adds constant-factor compute and, being a bottleneck, cannot exceed $\sim P_r$ simultaneously retrievable bindings. We report negative results (e.g., tasks where memory does not help) rather than omitting them, because the value of the claim rests on *where* and *why* the mechanism helps, established by controlled ablation—not on a single headline number.

VI. CONCLUSION

Mnemosyne shows that a small, linear-cost associative memory—persistent slots for universal motifs and register slots for per-sequence recall—restores the content-addressable, distance-independent retrieval that fixed-state SSMs lack, while preserving $O(L)$ scaling and exact RC symmetry. The next step is scale: the same layer, trained on genome-scale corpora, is the test of whether the mechanism translates from MQAR into biology.

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